LAB 1
DEPLOYING DVWA WEBSITE ON THE LINUX SERVERS

Objective

Students will learn to:
- Install and configure a LAMP stack (Linux, Apache, MySQL/MariaDB, PHP)
- Download and deploy DVWA
- Configure required permissions and database settings
- Test and verify DVWA deployment

Prerequisites

- A working Ubuntu Server (20.04+ recommended)
- Internet access
- Root or sudo privileges

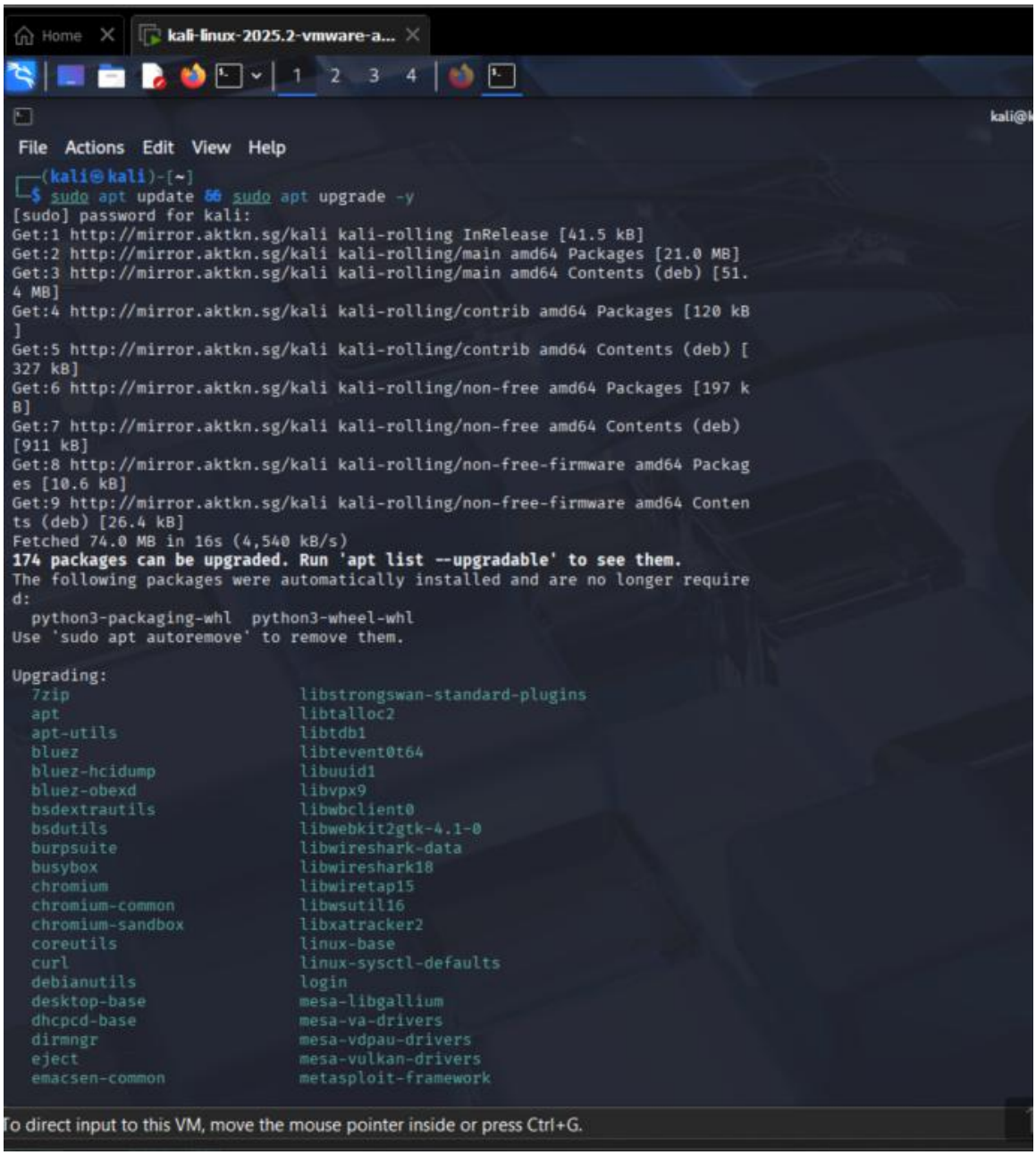
Note: Depending on the Linux distribution, the command line may be different.

PART 1. Deploying the website on a server

Step 1: Update the Server

sudo apt update && sudo apt upgrade -y

This command downloads the latest software package versions installed in the repositories and updates all currently installed packages to their latest versions. The -y option will automatically accept any prompt, so it is faster.



Step 2: Install Apache, MySQL, and PHP (LAMP Stack)

sudo apt install apache2 mysql-server php php-mysqli php-gd libapache2-mod-php git -y

sudo systemctl enable apache2
sudo systemctl start apache2

sudo systemctl status apache2

This will install MariaDB (in place of MySQL on Kali Linux) and major components for the LAMP stack, including Apache, PHP, and dependencies for the hosting of web applications.

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install apache2 -y
sudo systemctl enable apache2
sudo systemctl start apache2
apache2 is already the newest version (2.4.63-1).
apache2 set to manually installed.
The following packages were automatically installed and are no longer require
d:
  python3-packaging-whl  python3-wheel-whl
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
Synchronizing state of apache2.service with SysV service script with /usr/lib
/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service'
 → '/usr/lib/systemd/system/apache2.service'.

┌──(kali㉿kali)-[~]
└─$ sudo apt install mariadb-server -y
sudo systemctl enable mariadb
sudo systemctl start mariadb
mariadb-server is already the newest version (1:11.8.1-4).
mariadb-server set to manually installed.
The following packages were automatically installed and are no longer require
d:
  python3-packaging-whl  python3-wheel-whl
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
Synchronizing state of mariadb.service with SysV service script with /usr/lib
/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mariadb
Created symlink '/etc/systemd/system/multi-user.target.wants/mariadb.service'
 → '/usr/lib/systemd/system/mariadb.service'.
```

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl status mariadb

┌──(kali㉿kali)-[~]
└─$ sudo apt install php php-mysql php-gd libapache2-mod-php -y
php is already the newest version (2:8.4+96).
php set to manually installed.
php-mysql is already the newest version (2:8.4+96).
php-mysql set to manually installed.
libapache2-mod-php is already the newest version (2:8.4+96).
libapache2-mod-php set to manually installed.
The following packages were automatically installed and are no longer require
d:
  python3-packaging-whl  python3-wheel-whl
Use 'sudo apt autoremove' to remove them.

Installing:
  php-gd

Installing dependencies:
  php8.4-gd

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 40.5 kB
  Space needed: 190 kB / 63.7 GB available

Get:1 http://mirror.sg.gs/kali kali-rolling/main amd64 php8.4-gd amd64 8.4.8-
1 [36.6 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 php-gd all 2:8.4+96 [
3,960 B]
Fetched 40.5 kB in 1s (38.3 kB/s)
Selecting previously unselected package php8.4-gd.
(Reading database ... 413000 files and directories currently installed.)
Preparing to unpack .../php8.4-gd_8.4.8-1_amd64.deb ...
Unpacking php8.4-gd (8.4.8-1) ...
Selecting previously unselected package php-gd.
Preparing to unpack .../php-gd_2%3a8.4+96_all.deb ...
Unpacking php-gd (2:8.4+96) ...
Setting up php8.4-gd (8.4.8-1) ...
Creating config file /etc/php/8.4/mods-available/gd.ini with new version
Setting up php-gd (2:8.4+96) ...

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

Step 3: Clone the DVWA Repository

cd /var/www/html
sudo git clone https://github.com/digininja/DVWA.git

sudo chown -R www-data:www-data /var/www/html/DVWA
sudo chmod -R 755 /var/www/html/DVWA

The procedure involves changing directory to the web server root directory (/var/www/html), cloning the DVWA repository from GitHub, setting the www-data user as the owner so that the web server can access it, and setting permissions to 755 so that the server can read and run the files necessary for DVWA to run properly.

```
kali@kali: /var/www/html
File  Actions  Edit  View  Help
Setting up php-phpmyadmin-motranslator (5.4.0-1) ...
Setting up phpmyadmin (4:5.2.2-really+dfsg-1) ...
Determining localhost credentials from /etc/mysql/debian.cnf: succeeded.
dbconfig-common: writing config to /etc/dbconfig-common/phpmyadmin.conf
Creating config file /etc/dbconfig-common/phpmyadmin.conf with new version
Creating config file /etc/phpmyadmin/config-db.php with new version
checking privileges on database phpmyadmin for phpmyadmin@localhost: user creation needed.
granting access to database phpmyadmin for phpmyadmin@localhost: success.
verifying access for phpmyadmin@localhost: success.
creating database phpmyadmin: success.
verifying database phpmyadmin exists: success.
populating database via sql ...  done.
dbconfig-common: flushing administrative password
Processing triggers for libapache2-mod-php8.4 (8.4.8-1) ...
Processing triggers for kali-menu (2025.3.0) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for doc-base (0.11.2) ...
Processing 41 changed doc-base files, 6 added doc-base files...
Processing triggers for libc-bin (2.41-6) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for php8.4-cli (8.4.8-1) ...

┌──(kali㉿kali)-[~]
└─$ cd /var/www/html
sudo git clone https://github.com/digininja/DVWA.git

sudo chown -R www-data:www-data /var/www/html/DVWA
sudo chmod -R 755 /var/www/html/DVWA
Cloning into 'DVWA'...
remote: Enumerating objects: 5373, done.
remote: Counting objects: 100% (2206/2206), done.
remote: Compressing objects: 100% (719/719), done.
remote: Total 5373 (delta 1606), reused 1491 (delta 1487), pack-reused 3167 (from 1)
Receiving objects: 100% (5373/5373), 2.59 MiB | 472.00 KiB/s, done.
Resolving deltas: 100% (2735/2735), done.

┌──(kali㉿kali)-[/var/www/html]
└─$
```

Step 4: Configure DVWA

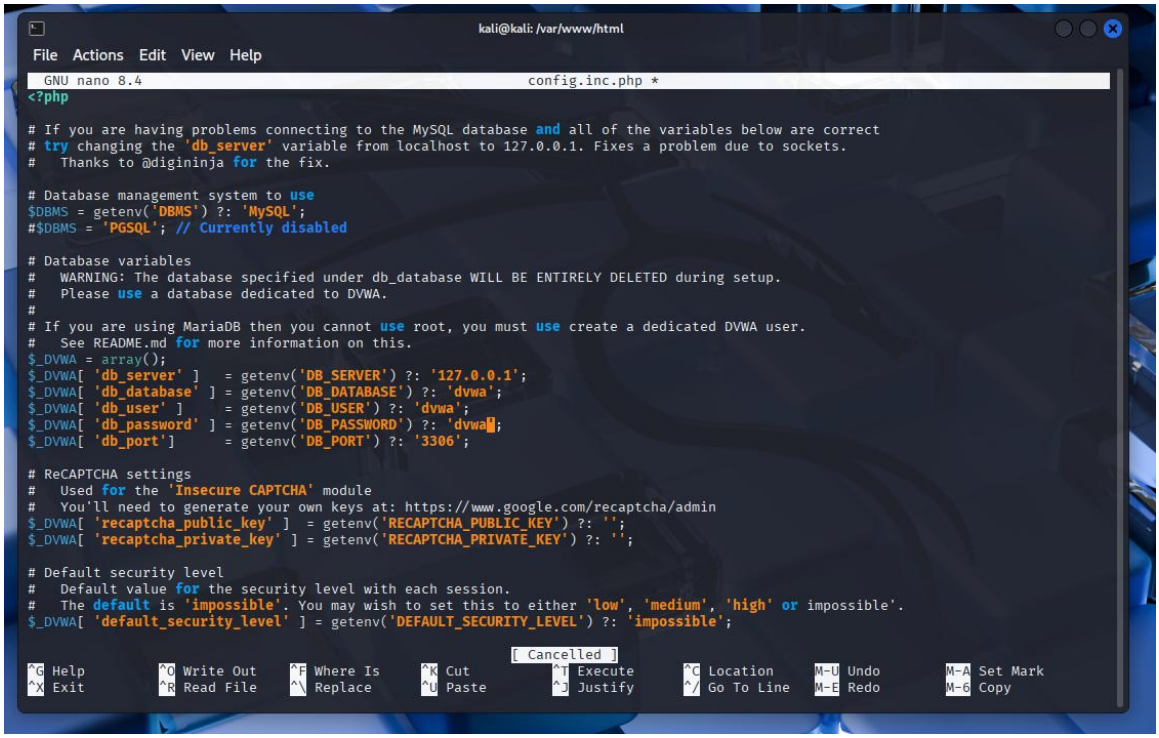
cd /var/www/html/DVWA/config

sudo cp config.inc.php.dist config.inc.php

sudo nano config.inc.php

```
kali@kali: /var/www/html
File  Actions  Edit  View  Help
  GNU nano 8.4                                   config.inc.php
<?php

# If you are having problems connecting to the MySQL database and all of the variables below are correct
# try changing the 'db_server' variable from localhost to 127.0.0.1. Fixes a problem due to sockets.
#   Thanks to @digininja for the fix.

# Database management system to use
$DBMS = getenv('DBMS') ?: 'MySQL';
#$DBMS = 'PGSQL'; // Currently disabled

# Database variables
#   WARNING: The database specified under db_database WILL BE ENTIRELY DELETED during setup.
#   Please use a database dedicated to DVWA.
#
# If you are using MariaDB then you cannot use root, you must use create a dedicated DVWA user.
#   See README.md for more information on this.
$_DVWA = array();
$_DVWA[ 'db_server' ]   = getenv('DB_SERVER') ?: '127.0.0.1';
$_DVWA[ 'db_database' ] = getenv('DB_DATABASE') ?: 'dvwa';
$_DVWA[ 'db_user' ]     = getenv('DB_USER') ?: 'dvwa';
$_DVWA[ 'db_password' ] = getenv('DB_PASSWORD') ?: 'p@ssw0rd';
$_DVWA[ 'db_port']      = getenv('DB_PORT') ?: '3306';

# ReCAPTCHA settings
#   Used for the 'Insecure CAPTCHA' module
#   You'll need to generate your own keys at: https://www.google.com/recaptcha/admin
$_DVWA[ 'recaptcha_public_key' ]  = getenv('RECAPTCHA_PUBLIC_KEY') ?: '';
$_DVWA[ 'recaptcha_private_key' ] = getenv('RECAPTCHA_PRIVATE_KEY') ?: '';

# Default security level
#   Default value for the security level with each session.
#   The default is 'impossible'. You may wish to set this to either 'low', 'medium', 'high' or impossible'.
$_DVWA[ 'default_security_level' ] = getenv('DEFAULT_SECURITY_LEVEL') ?: 'impossible';

                                [ Read 56 lines (converted from DOS format) ]
^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy
```

Edit the following lines:
$_DVWA['db_user'] = 'dvwa';
$_DVWA['db_password'] = 'dvwa';



This is to move into the DVWA configuration directory, copying the default configuration file (config.inc.php.dist) to get an editable version (config.inc.php), and modifying it using the nano text editor to place the database username and password as 'dvwa' to access the DVWA database.

Step 5: Create the DVWA Database

sudo mysql

Then run:
CREATE DATABASE dvwa;
CREATE USER 'dvwa'@'localhost' IDENTIFIED BY 'dvwa';
GRANT ALL PRIVILEGES ON dvwa.* TO 'dvwa'@'localhost';
FLUSH PRIVILEGES;
EXIT;

Open the MariaDB client, make a database dvwa, and create a user account 'dvwa' with password 'dvwa' that can connect from the localhost, making the database usable.

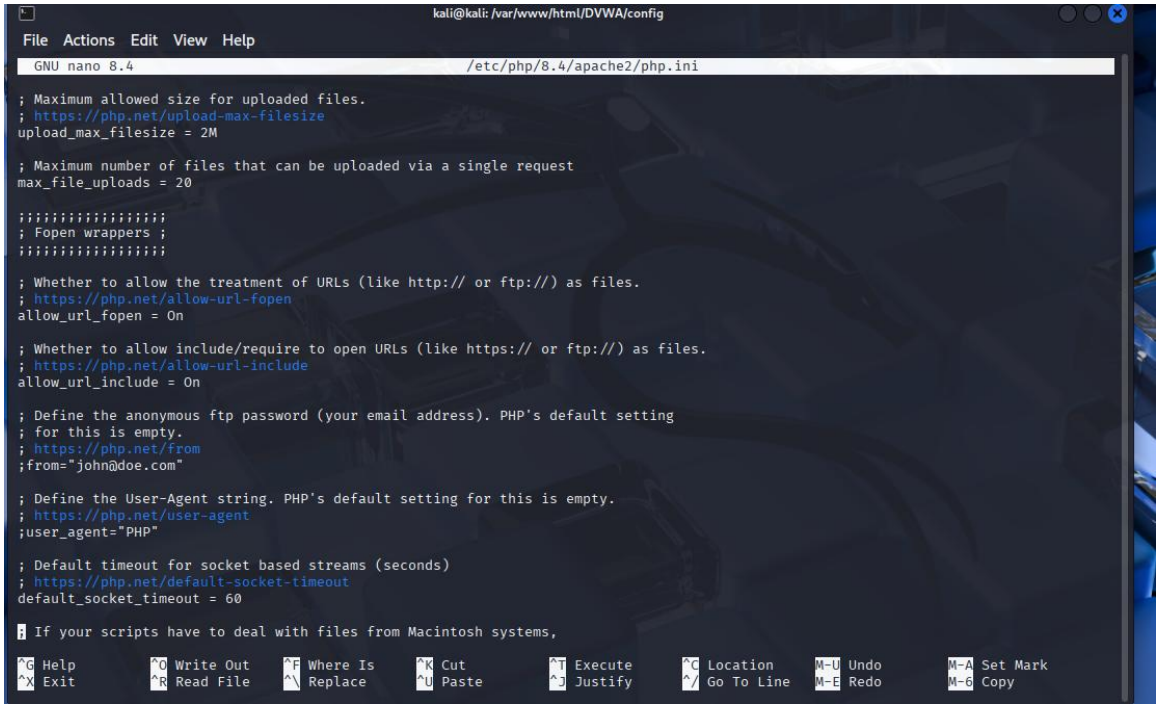
```
┌──(kali㉿kali)-[/var/www/html]
└─$ cd /var/www/html/DVWA/config
sudo cp config.inc.php.dist config.inc.php
sudo nano config.inc.php

┌──(kali㉿kali)-[/var/www/html/DVWA/config]
└─$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 43
Server version: 11.8.1-MariaDB-4 Debian n/a

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Support MariaDB developers by giving a star at https://github.com/MariaDB/server
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE dvwa;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> CREATE USER 'dvwa'@'localhost' IDENTIFIED BY 'dvwa';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON dvwa.* TO 'dvwa'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> EXIT;
Bye

┌──(kali㉿kali)-[/var/www/html/DVWA/config]
└─$
```

Step 6: Configure PHP

sudo nano /etc/php/*/apache2/php.ini

```
kali@kali: /var/www/html/DVWA/config
File  Actions  Edit  View  Help
  GNU nano 8.4                    /etc/php/8.4/apache2/php.ini
[PHP]

;;;;;;;;;;;;;;;;;;;
; About php.ini   ;
;;;;;;;;;;;;;;;;;;;
; PHP's initialization file, generally called php.ini, is responsible for
; configuring many of the aspects of PHP's behavior.

; PHP attempts to find and load this configuration from a number of locations.
; The following is a summary of its search order:
; 1. SAPI module specific location.
; 2. The PHPRC environment variable.
; 3. A number of predefined registry keys on Windows
; 4. Current working directory (except CLI)
; 5. The web server's directory (for SAPI modules), or directory of PHP
; (otherwise in Windows)
; 6. The directory from the --with-config-file-path compile time option, or the
; Windows directory (usually C:\windows)
; See the PHP docs for more specific information.
; https://php.net/configuration.file

; The syntax of the file is extremely simple.  Whitespace and lines
; beginning with a semicolon are silently ignored (as you probably guessed).
; Section headers (e.g. [Foo]) are also silently ignored, even though
; they might mean something in the future.

; Directives following the section heading [PATH=/www/mysite] only
; apply to PHP files in the /www/mysite directory.  Directives
; following the section heading [HOST=www.example.com] only apply to
; PHP files served from www.example.com.  Directives set in these
; special sections cannot be overridden by user-defined INI files or
; at runtime. Currently, [PATH=] and [HOST=] sections only work under
; CGI/FastCGI.
; https://php.net/ini.sections
                                        [ Read 1855 lines ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy
```

Find and set:
allow_url_include = On

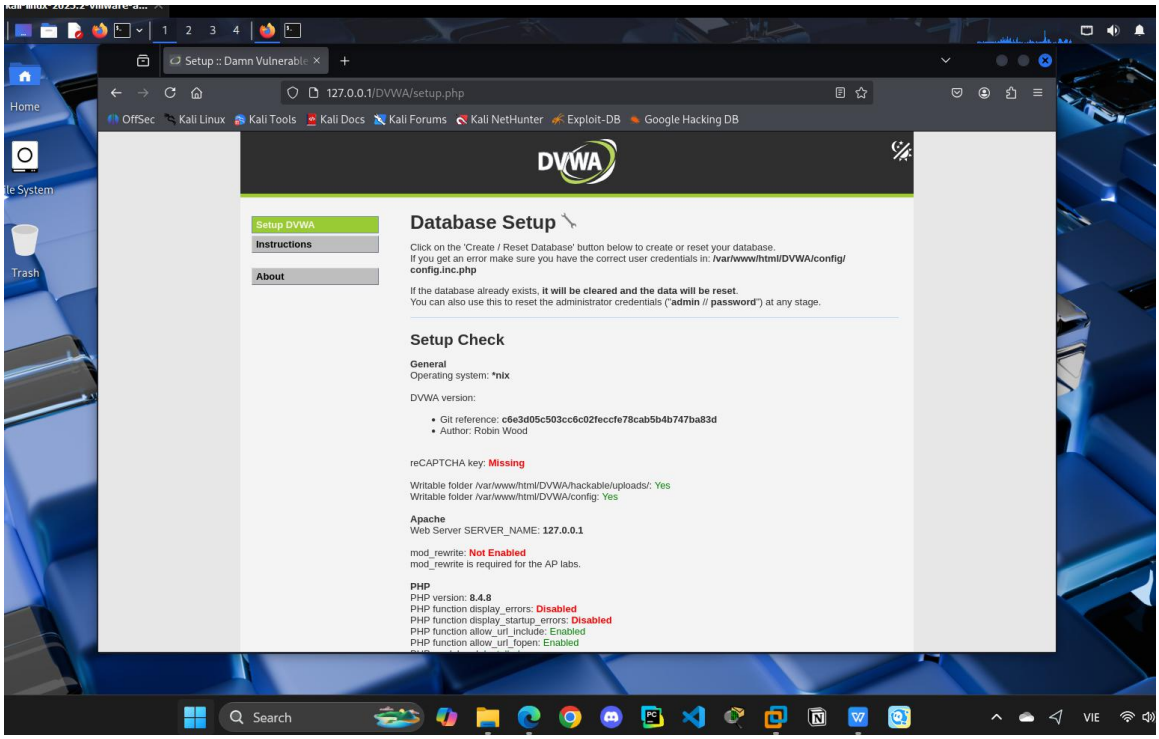


Then restart Apache:
sudo systemctl restart apache2

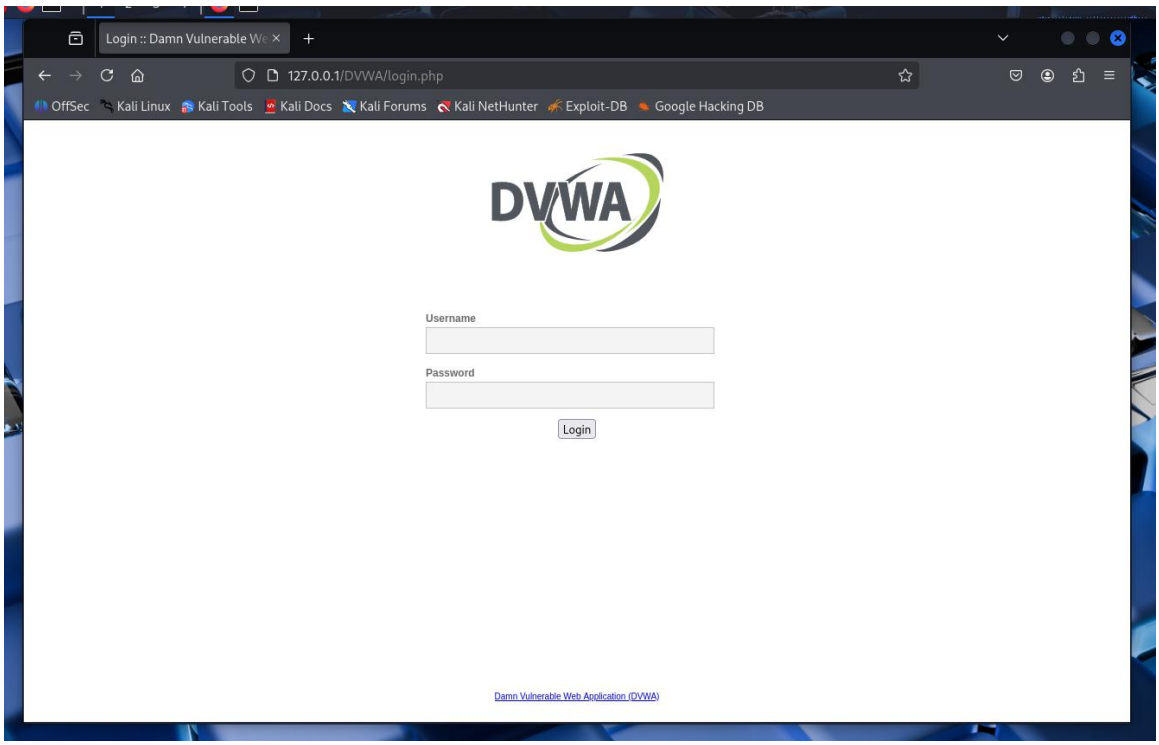
Make changes to the PHP configuration file to enable the allow_url_include option, allowing the web server to include URLs from outside, which DVWA requires so that it can be used.

Step 7: Access DVWA

Visit: http://<your-server-ip>/DVWA/setup.php



Click 'Create / Reset Database'

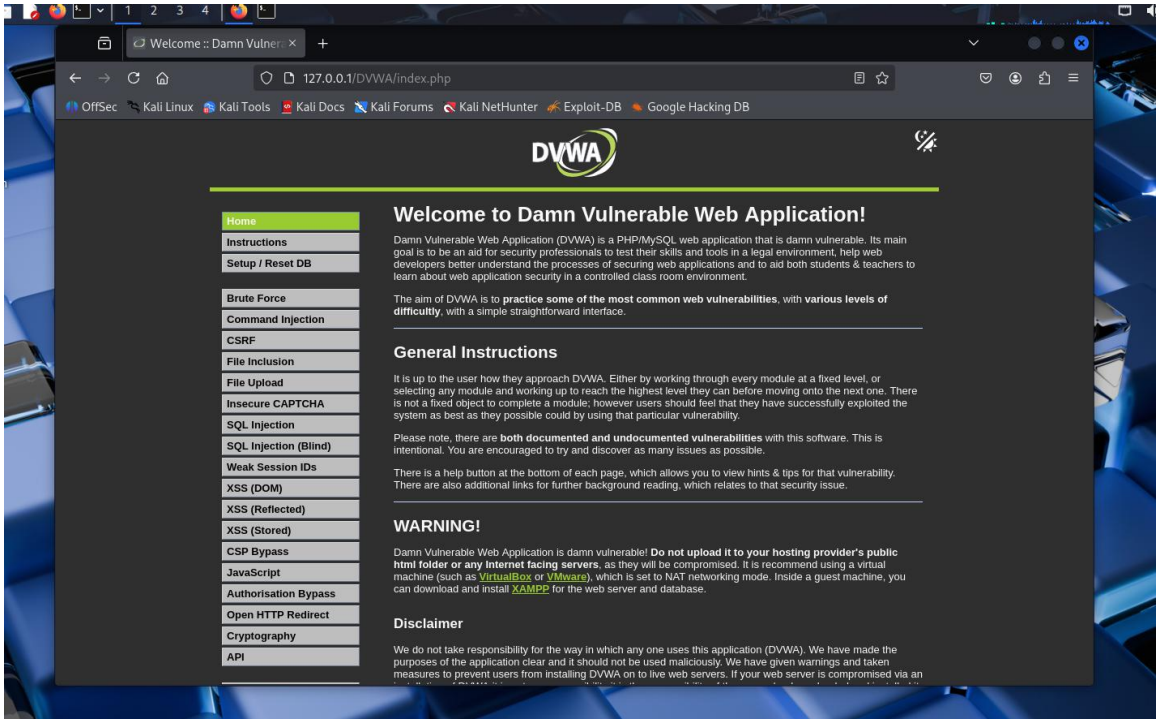


Then go to: http://<your-server-ip>/DVWA/login.php

Login with:

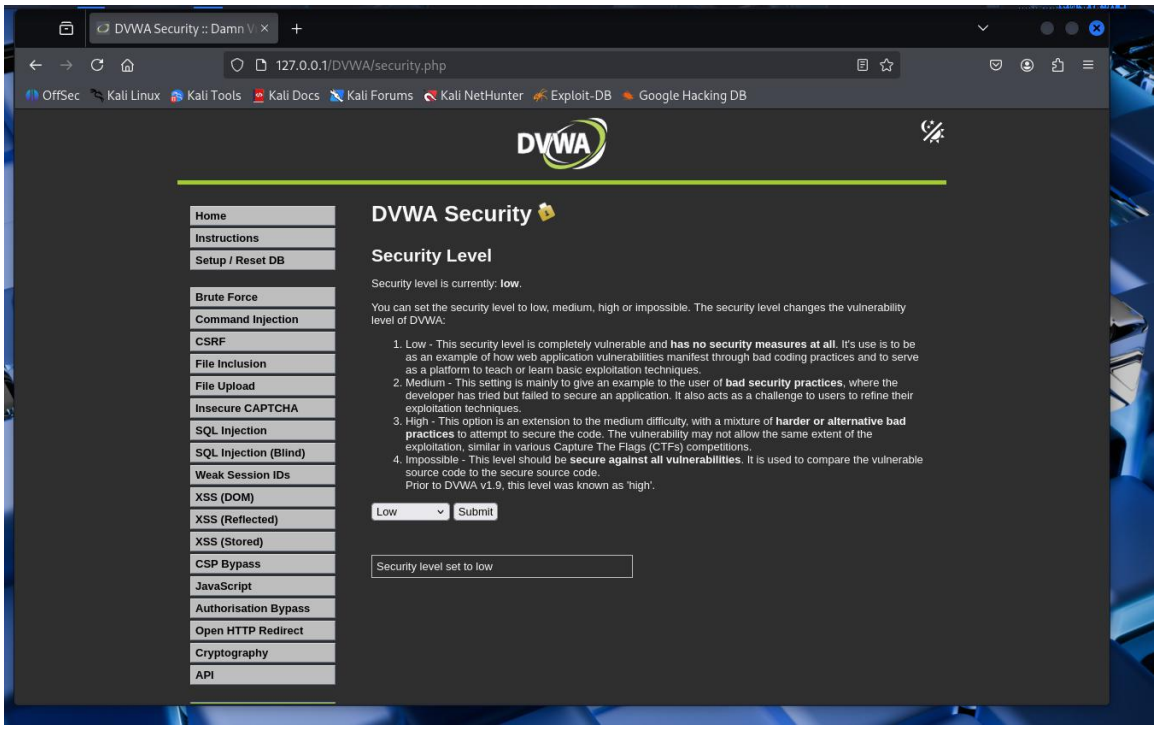
Username: admin

Password: password



Step 8: Verification & Tasks

- Screenshot the DVWA login page
- Log in and set security level to 'low'
- Write a short reflection on why DVWA is insecure by design



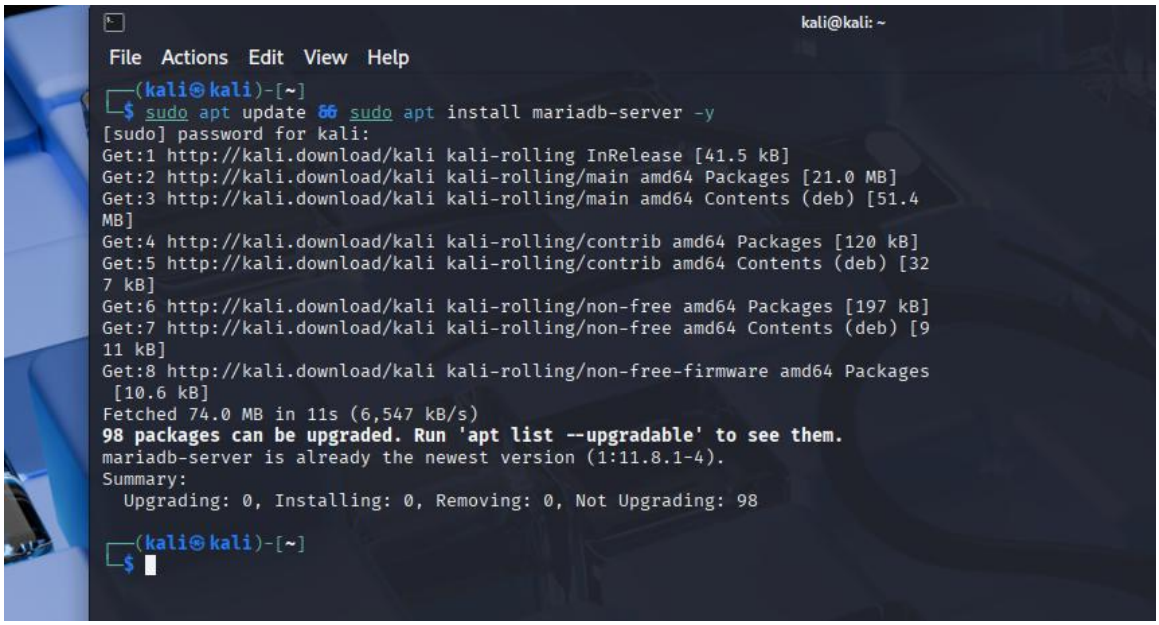
PART 2. Deploying the website on 2 servers

Network Configuration

Example configuration (Students need to check the IP address of the Server):
- Web Server: 192.168.1.20 -> 192.168.88.129 (hostname -I)
- Database Server: 192.168.1.30 -> 192.168.88.130

Step 1: Configure Database Server

1. Update packages and install MariaDB:
   sudo apt update && sudo apt install mariadb-server -y



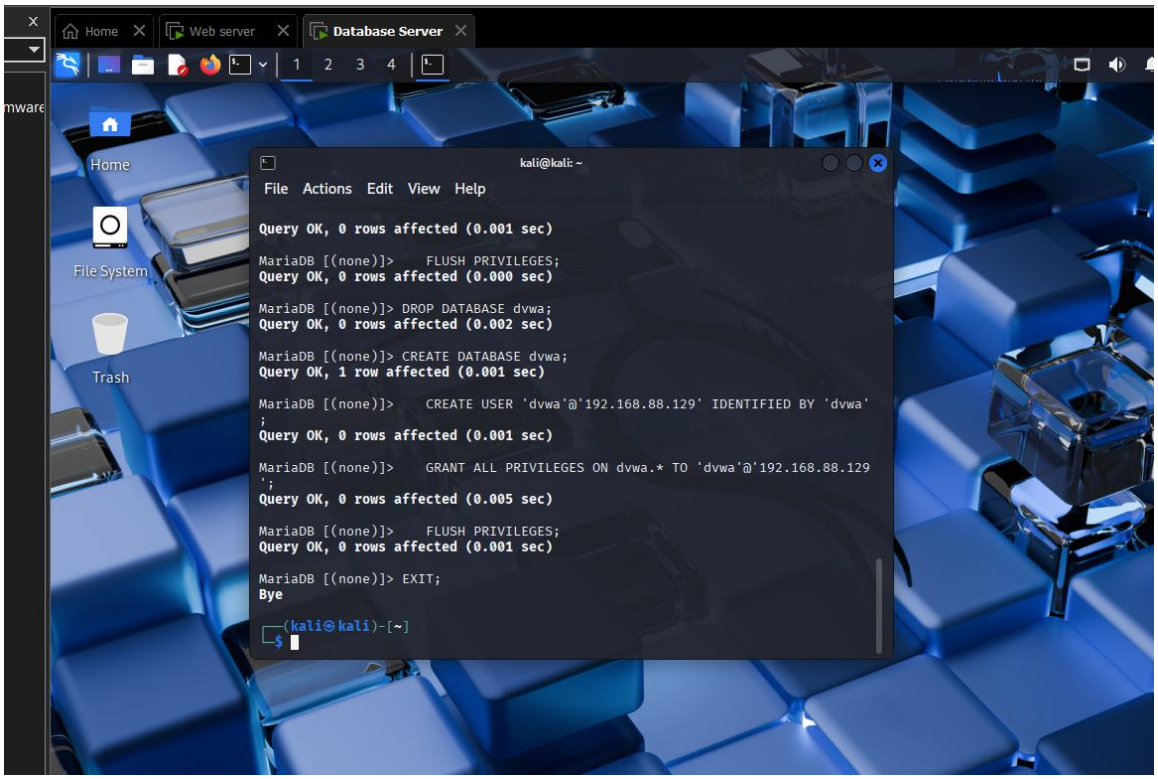
This step updates the package index in Kali Linux with sudo apt update and installs the MariaDB server package with sudo apt install mariadb-server -y. This installs a database server necessary to handle and store data for applications such as DVWA.

2. Start and enable MariaDB:
   sudo systemctl enable mariadb
   sudo systemctl start mariadb

Configures the MariaDB service to start automatically on system boot and starts it right away to ensure the database is running and available for use.

3. Configure the DVWA database:
   sudo mysql

   CREATE DATABASE dvwa;
   CREATE USER 'dvwa'@'192.168.88.129' IDENTIFIED BY 'dvwa';
   GRANT ALL PRIVILEGES ON dvwa.* TO 'dvwa'@'192.168.88.129';
   FLUSH PRIVILEGES;
   EXIT;



This command removes any existing database and user, adds a new user 'dvwa' with password 'dvwa' that can log in using IP address 192.168.88.129, grants the above user full rights to manage the dvwa database, flushes the privilege table to apply these changes, and closes the MariaDB interface.

4. Allow remote connections:
   sudo nano /etc/mysql/mariadb.conf.d/50-server.cnf
   Change: bind-address = 0.0.0.0

```
  GNU nano 8.4                                  /etc/mysql/mariadb.conf.d/50-server.cnf
#
# These groups are read by MariaDB server.
# Use it for options that only the server (but not clients) should see

# this is read by the standalone daemon and embedded servers
[server]

# this is only for the mariadbd daemon
[mariadbd]

#
# * Basic Settings
#

#user                    = mysql
pid-file                = /run/mysqld/mysqld.pid
basedir                 = /usr
#datadir                 = /var/lib/mysql
#tmpdir                  = /tmp

# Broken reverse DNS slows down connections considerably and name resolve is
# safe to skip if there are no "host by domain name" access grants
#skip-name-resolve

# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address            = 0.0.0.0

#
# * Fine Tuning
#

#key_buffer_size         = 128M
#max_allowed_packet      = 1G
#thread_stack            = 192K
#thread_cache_size       = 8
# This replaces the startup script and checks MyISAM tables if needed
                                                                  [ Read 117 lines ]
```

sudo systemctl restart mariadb

Edit the MariaDB configuration file using nano to bind-address to 0.0.0.0 to enable the database server to listen for connections on any IP address and reboot the MariaDB service to implement the changes

Step 2: Configure Web Server

1. Install Apache, PHP, and dependencies:

   sudo apt update

   sudo apt install apache2 php php-mysqli php-gd git -y

```
kali@kali: ~
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ sudo apt update
   sudo apt install apache2 php php-mysqli php-gd git -y
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [51.4
MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [120 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [32
7 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [197 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [9
11 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages
 [10.6 kB]
Fetched 74.0 MB in 11s (6,844 kB/s)
98 packages can be upgraded. Run 'apt list --upgradable' to see them.
Note, selecting 'php8.4-mysql' instead of 'php-mysqli'
apache2 is already the newest version (2.4.63-1).
php is already the newest version (2:8.4+96).
php8.4-mysql is already the newest version (8.4.8-1).
php-gd is already the newest version (2:8.4+96).
Upgrading:
  git  git-man

Summary:
  Upgrading: 2, Installing: 0, Removing: 0, Not Upgrading: 96
  Download size: 11.1 MB
  Space needed: 244 kB / 64.2 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 git-man all 1:2.47.2-0.2 [2,205 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 git amd64 1:2.47.2-0.2 [8,858 kB]
Fetched 11.1 MB in 2s (5,181 kB/s)
(Reading database ... 410847 files and directories currently installed.)
Preparing to unpack .../git-man_1%3a2.47.2-0.2_all.deb ...
Unpacking git-man (1:2.47.2-0.2) over (1:2.47.2-0.1) ...
Preparing to unpack .../git_1%3a2.47.2-0.2_amd64.deb ...
Unpacking git (1:2.47.2-0.2) over (1:2.47.2-0.1) ...
Setting up git-man (1:2.47.2-0.2) ...
Setting up git (1:2.47.2-0.2) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.3.0) ...

┌──(kali㉿kali)-[~]
```

Updates the list of packages and installs the Apache webserver, PHP, PHP-GD library, and Git, which provide the needed pieces to allow web-based applications like DVWA that rely on web server and PHP functionalities.

2. Clone DVWA into web root:
   cd /var/www/html
   sudo git clone https://github.com/digininja/DVWA.git

```
┌──(kali㉿kali)-[/var/www/html]
└─$ cd /var/www/html
   sudo git clone https://github.com/digininja/DVWA.git
Cloning into 'DVWA'...
remote: Enumerating objects: 5373, done.
remote: Counting objects: 100% (2183/2183), done.
remote: Compressing objects: 100% (709/709), done.
remote: Total 5373 (delta 1589), reused 1474 (delta 1474), pack-reused 3190 (from 2)
Receiving objects: 100% (5373/5373), 2.60 MiB | 10.46 MiB/s, done.
Resolving deltas: 100% (2730/2730), done.

┌──(kali㉿kali)-[/var/www/html]
└─$
```

Moves the DVWA repository from GitHub into the root directory of the web server (/var/www/html).

3. Set permissions:
   sudo chown -R www-data:www-data DVWA

sudo chmod -R 755 DVWA

```
┌──(kali㉿kali)-[/var/www/html]
└─$ sudo chown -R www-data:www-data DVWA
   sudo chmod -R 755 DVWA
```

Alters ownership of the DVWA directory and files to www-data group and user, and sets permissions to 755 so that owner has read, write, and execute capabilities, but group and others have read and execute.

4. Configure DVWA:

cd DVWA/config

sudo cp config.inc.php.dist config.inc.php

sudo nano config.inc.php

Set:

$_DVWA['db_server'] = '192.168.88.130';

$_DVWA['db_user'] = 'dvwa';

$_DVWA['db_password'] = 'dvwa';

$_DVWA['db_database'] = 'dvwa';

```
kali@kali: /var/www/html/DVWA/config
File  Actions  Edit  View  Help
  GNU nano 8.4                                   config.inc.php *
<?php

# If you are having problems connecting to the MySQL database and all of the variables below are correct
# try changing the 'db_server' variable from localhost to 127.0.0.1. Fixes a problem due to sockets.
#   Thanks to @digininja for the fix.

# Database management system to use
$DBMS = getenv('DBMS') ?: 'MySQL';
#$DBMS = 'PGSQL'; // Currently disabled

# Database variables
#   WARNING: The database specified under db_database WILL BE ENTIRELY DELETED during setup.
#   Please use a database dedicated to DVWA.
#
# If you are using MariaDB then you cannot use root, you must use create a dedicated DVWA user.
#   See README.md for more information on this.
$_DVWA = array();
$_DVWA[ 'db_server' ]   = getenv('DB_SERVER') ?: '192.168.1.30';
$_DVWA[ 'db_database' ] = getenv('DB_DATABASE') ?: 'dvwa';
$_DVWA[ 'db_user' ]     = getenv('DB_USER') ?: 'dvwa';
$_DVWA[ 'db_password' ] = getenv('DB_PASSWORD') ?: 'dvwa';
$_DVWA[ 'db_port']      = getenv('DB_PORT') ?: '3306';

# ReCAPTCHA settings
#   Used for the 'Insecure CAPTCHA' module
#   You'll need to generate your own keys at: https://www.google.com/recaptcha/admin
$_DVWA[ 'recaptcha_public_key' ]  = getenv('RECAPTCHA_PUBLIC_KEY') ?: '';
$_DVWA[ 'recaptcha_private_key' ] = getenv('RECAPTCHA_PRIVATE_KEY') ?: '';

# Default security level
#   Default value for the security level with each session.
#   The default is 'impossible'. You may wish to set this to either 'low', 'medium', 'high' or impossible'.
$_DVWA[ 'default_security_level' ] = getenv('DEFAULT_SECURITY_LEVEL') ?: 'impossible';

# Default locale
#   Default locale for the help page shown with each session.
#   The default is 'en'. You may wish to set this to either 'en' or 'zh'.
$_DVWA[ 'default_locale' ] = getenv('DEFAULT_LOCALE') ?: 'en';

# Disable authentication
#   Some tools don't like working with authentication and passing cookies around

^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-] To Bracket
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy      ^B Where Was
```

Ensure 'allow_url_include = On' in PHP config:

sudo nano /etc/php/8.1/apache2/php.ini

Then restart Apache:

sudo systemctl restart apache2

Before step 3 run:
sudo apt install ufw

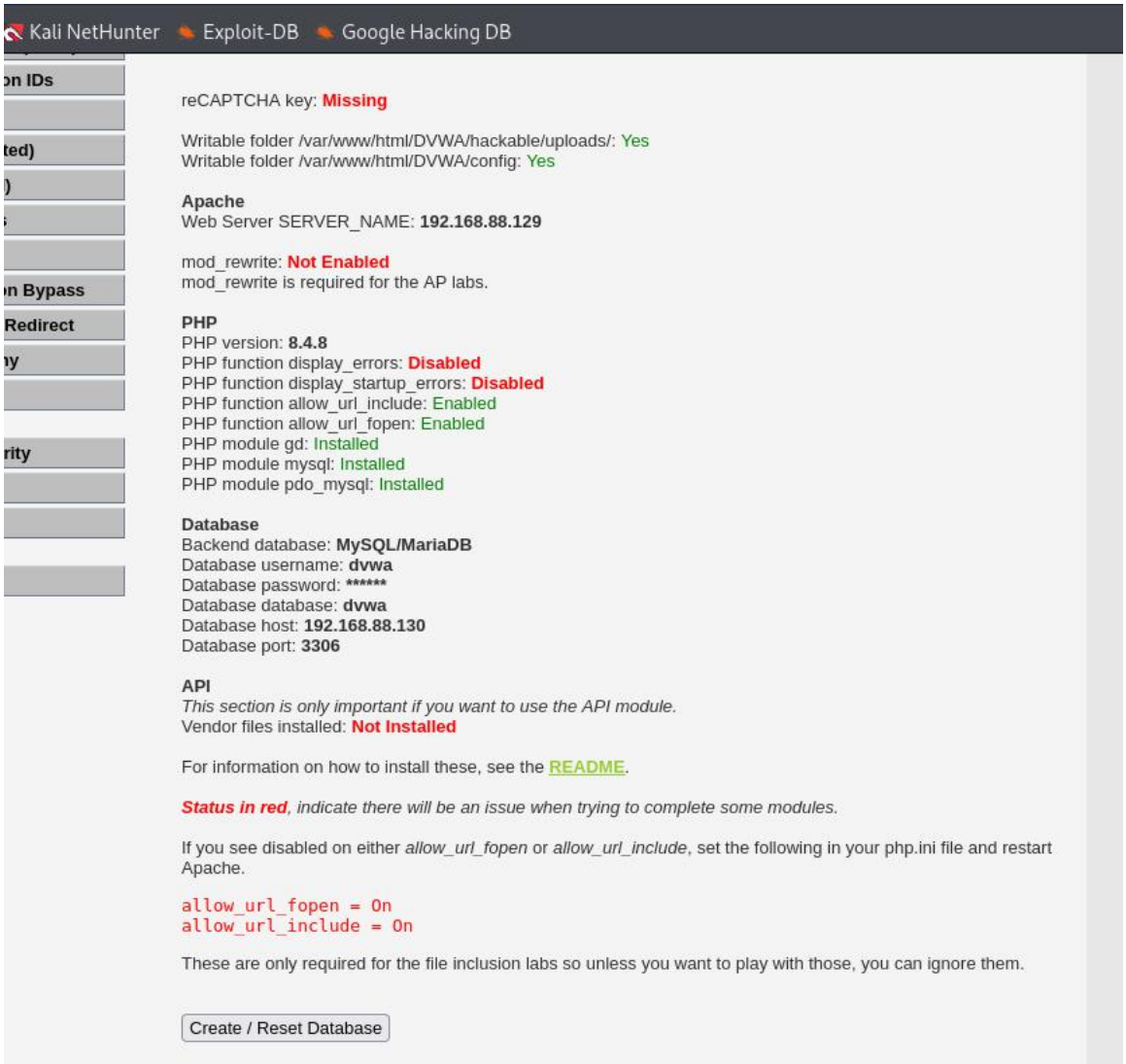
sudo ufw status

sudo ufw disable

Enters into the DVWA configuration directory, copies the default configuration file for editing without altering the original one and modifies the copy by adding the database connection data: IP address of 192.168.88.130, username 'dvwa', password 'dvwa', database name 'dvwa'. And also modifies PHP configuration by setting the allow_url_include option to enable it and restarts the Apache server to ensure that DVWA executes successfully with no issues.

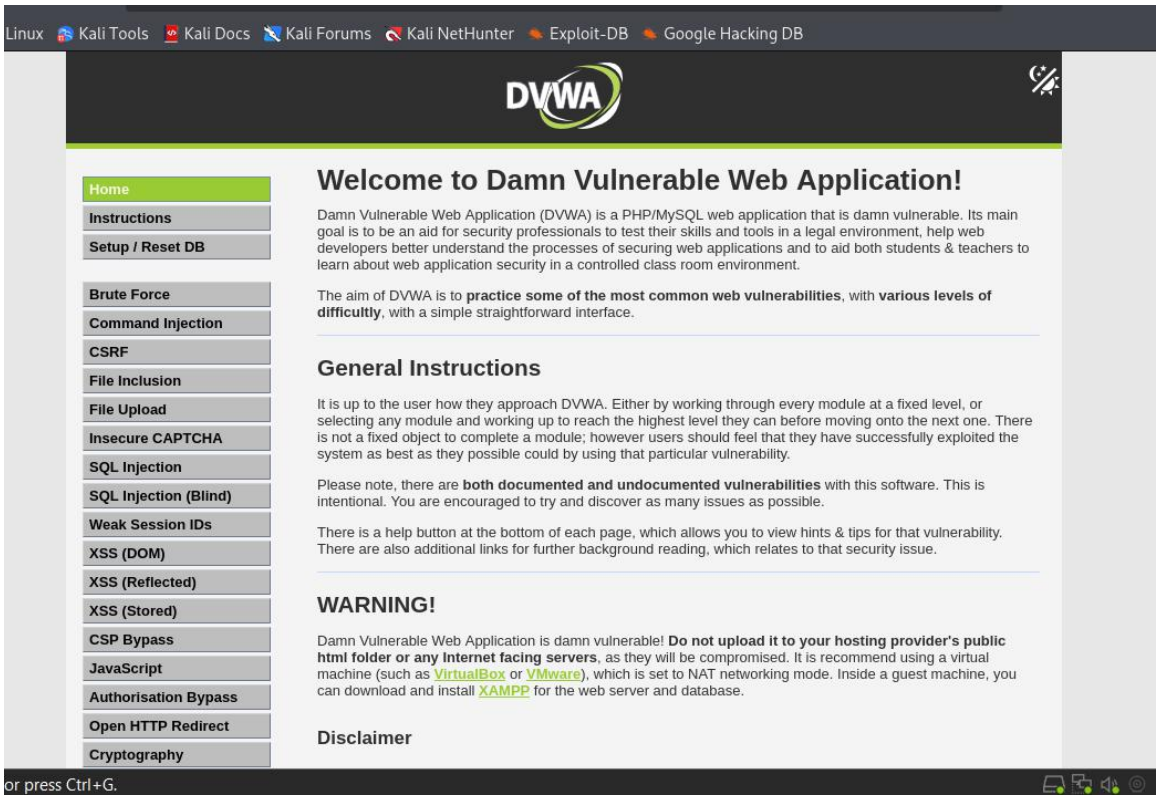
Step 3: Verify Setup

1. On the Web Server, open browser or client and visit:
   http://192.168.1.20/DVWA/setup.php -> http://192.168.88.129/DVWA/setup.php



2. Click 'Create / Reset Database'
3. Then login at:
   http://192.168.1.20/DVWA/login.php -> http://192.168.88.129/DVWA/login.php

   Username: admin
   Password: password



Common Troubleshooting Tips

- Ensure MySQL is listening on 0.0.0.0 s
- Use `telnet 192.168.1.30 3306` from the Web Server to test DB connection
- Check firewall (ufw or iptables) if services are blocked
- Check Apache and MariaDB logs for error details

Lab Report Submission

Students must submit:
- Screenshot of MySQL configuration and DVWA login page
- Evidence of IP setup and working DB connectivity
- A short explanation of each configuration step